Data cleaning and preparation

How the tables behind the stand's models and maps were produced from XRF field measurements and chemicals.tj portal materials. For each step: what was checked, what was fixed, and which script reproduces it.

> Common rules for both topics: source files are never modified; every value in the final tables refers to the document or measurement it came from; disputed cases are not deleted but flagged and kept in the data; all steps are performed by scripts and can be rerun from the source materials.

1. Heavy metals (XRF)

1.1. Source data

- Field measurements with a portable X-ray fluorescence analyser (XRF): 10 warehouses, 169 points, 9 elements (As, Co, Hg, Mo, Pb, Rb, Th, U, Zn).
- Warehouse outlines (polygons) and measurement point coordinates in GeoJSON.
- Open data for model features: rivers and canals (OpenStreetMap), soil pH (SoilGrids), terrain, climate, population, land use, geology.

1.2. Loading and fixed format problems

Problem	Symptom	Solution
Cyrillic file names	different Unicode normalisation on macOS (NFC/NFD): files were silently not found	files renamed to Latin characters
Duplicate `name` column in GeoJSON	name conflict when renaming fields	only the required columns are selected before renaming
Three-dimensional coordinates (EPSG:4979, Z = 0)	mismatch with the database coordinate system (EPSG:4326)	coordinates are converted to 2D before writing
The Kushoniyon warehouse outline duplicated the Shahrinav outline	found when cross-checking geometries at an early stage	outline replaced; the 50 and 100 m buffers are built from the outline automatically (a database view) rather than stored separately, so the geometries cannot diverge

Loading script: `db/import_data.py`.

1.3. Geometry checks

- All 10 outlines were checked for validity and self-intersections, and all 45 pairs of outlines for overlaps and buffer nesting (`db/qc_geometry.py`). No issues found.
- Warehouse areas range from 0.05 to 1.07 ha.
- 49 of 169 points lie inside an outline; for points outside, the median distance to the outline is 49 m, 11 points are more than 100 m away, the farthest 126 m. Points outside the outline are not removed: the signed distance to the outline (inside/outside) and the zone (inside, up to 50 m, up to 100 m, beyond) are used as model features.

1.4. Values

- Values below the detection limit are not discarded: they are replaced with half the limit (LOD/2) and flagged `below_lod`.
- The model uses the logarithm of concentration minus the median logarithm of its warehouse. The model thus learns the position of the hotspot within the warehouse, not differences between warehouses; with 10 warehouses an absolute country-wide prediction is not feasible.
- For portable XRF the detection limits for arsenic and mercury are close to the standards, so exceedances of As and Hg are treated as a signal for laboratory verification. XRF is reliable for lead and zinc.

1.5. Open-data features

All candidates are collected and stored in the database: distance to water, pH, slope, elevation, population, land use, geology, flow accumulation, climate. Each was tested with the same scheme: training on 9 warehouses, testing on the tenth (LOWO). Only features that do not degrade performance in combination entered the model: distance to water per point (OpenStreetMap) and soil pH per warehouse (SoilGrids). One warehouse has no pH value; the sample mean is used. Details and test results are on the "Methodology: metals" page.

Scripts: `db/fetch_point_covariates.py`, `db/fetch_landcover_population.py`, `db/diagnose_features.py`, `db/scan_all_substances.py`.

1.6. Derived tables

- Warehouse priority: `db/rank_sites_priority.py`, a composite index over Pb, As, Hg, Zn (log excess over the median of the whole survey).
- Contamination zones within a warehouse: `db/build_risk_zones.py`, from the actual points, without interpolation.
- Pb model: `db/run_model_and_store.py`.
- Export for notebooks without a database: `db/export_geopackage.py` → `tajikistan_soil_ml_export.gpkg`.

2. POPs pesticide sites (chemicals.tj)

2.1. Source data

The client has no structured tables. All information comes from the public part of the chemicals.tj portal:

- the map: 192 sites of type "POPs" with coordinates and status ("contaminated" / "cleaned up");
- a survey form for each site with about 50 fields: area, odour, pit, access, water, groundwater, residents in 50/100/500 m rings, etc.;
- a text description of the site;
- attachments: photos, laboratory protocols, sampling logs and registers, reports.

Export script: `db/fetch_chemicals_tj.py --download-docs`.

2.2. Documents

- The same documents are attached to different sites. Copies were merged by file checksum and by matching extracted text: 245 distinct documents remained out of 373 attachments.
- Documents were classified as laboratory protocol, sampling log or register, report, or other. Whether the PDF has a text layer was recorded.

2.3. Extraction of analyses

Document type	How it is extracted	Check
Text PDFs (AFI protocol No. 8, ILIM report No. 42, Institute of Chemistry table of 2012, logs)	table parsing by templates; logs come in several templates, including blocks covering several sites	every value was found verbatim in the text of its document; 32 cases where figures overlap with text from a neighbouring column are flagged
Scans without a text layer (AFI protocol No. 126 with 286 samples; Institute of Chemistry letter of 2017)	transcribed manually into `manual/*.csv` tables with notes on typos in the original	the 2019 logs matched protocol No. 126 for 229 of 233 "site × substance" sets

Script: `db/extract_pop_tables.py` → `extracted/measurements_long.csv`.

2.4. Harmonisation

- The unit is mg/kg. Substance names are mapped to a single dictionary: DDT, DDD, DDE, HCH isomers (α, β, γ, δ), aldrin, dieldrin, heptachlor, etc.
- "Not detected" is stored as a separate flag and counts as zero in group totals.

- Log columns without a substance name are kept in separate fields and do not enter the contamination indicator.
- The site contamination indicator is the maximum over samples of two totals: DDT+DDD+DDE and total HCH isomers.

> A fixed build error: if one substance of a group was "not detected" in a sample, the group total became empty, and some sites lost their contamination value. After the fix the number of sites with analyses rose from 159 to 169; the model, risk index, sampling plan and audit were recalculated.

2.5. Duplicates between sources

The same sample often appears both in a sampling log and in a laboratory protocol. Each value is assigned a source rank:

Rank	Source	In the consolidated sample
1	laboratory protocol	yes
2	log or register, if the sample is not in any protocol	yes
0	log duplicating a protocol	no, kept in the data with a flag

Where values differ, the protocol goes into the consolidated sample. The 2020 registers matched protocol No. 8 for 47 of 48 sites. Result: 586 samples and 2823 concentration values without duplicates; 4572 values were extracted in total.

2.6. Sample coordinates

Laboratory protocols contain no coordinates. Coordinates were transferred from logs and registers of the same site, and the method is recorded for each sample:

- by matching the DDT value in the log and the protocol (protocol No. 126);
- by the order of replicates T1/T2/K3 in the register (protocol No. 8);
- by sample number (ILIM report, 2016);
- from the log itself, if it gives coordinates.

This gave coordinates for 462 samples; the median distance from a sample to the site point is about 30 m. Samples without coordinates are placed at the site point with a flag; sites that are not on the map remain without geometry.

2.7. Site code matching

Protocol No. 126 records 12 sites under field codes tj77xx that do not exist on the map. Nine of them were confidently linked to map cards by village name (table

`manual/protocol126_map_links.csv`); 3 remain ambiguous and are kept in the layer as separate sites without coordinates. Code TJ-7580 is used twice on the map for different cards; the sites differ by coordinates.

2.8. Information from card text

About 2200 facts were extracted from text descriptions by patterns: distances to housing, water, roads, crops, a school and a well; numbers of residents, children, families and houses; quantity of pesticides, area; mentions of children, livestock, crops, drinking water, health complaints, remediation. For each fact the source text fragment is kept so that it can be checked.

2.9. Open-data features

For each site the following were computed from local rasters (no internet requests): soil pH (SoilGrids), elevation and slope (150 m terrain), population within ~500 m and 1 km (WorldPop 2020), flow accumulation, mean annual temperature and precipitation (WorldClim 2.1). Script: `db/pop_site_covariates.py`.

2.10. Audit of portal data quality

As a by-product of the preparation, a list of problems in the portal itself was compiled, with evidence and the action needed to fix each (`audit/data_quality_audit.csv`, script `db/pop_data_audit.py`): 63 problems, 16 of them of high importance.

Category	Problems	Examples
Identification	19	repeated code TJ-7580; protocol codes not on the map; near-duplicate sites
Coordinates	13	samples more than 2 km from the site point; swapped latitude and longitude headers; low precision
Analyses	10	discrepancies between log and protocol; columns without a substance name
Status	9	six sites are marked "cleaned up", but analyses show contamination above the standard
Documents	7	logs attached to other sites; scans without a text layer
Other	5	different log templates, empty survey-form fields, service text in descriptions

2.11. Result

The layer `source_data/chemicals_tj/pop_layer/pop_layer.gpkg` and the same tables as CSV; field descriptions in `data_dictionary.md` (in Russian).

Layer	Contents
`pop_sites`	sites: survey form, contamination summary, facts from text, links to documents
`pop_samples`	samples of the consolidated set, substances in separate columns, coordinate source
`pop_measurements`	all extracted "sample × substance" values, including duplicates (rank 0)
`pop_text_facts`	fact from text and its source fragment
`pop_files`	all site attachments with links
`pop_sources`	sources of laboratory data

3. How to reproduce

Metals (requires the stand's PostgreSQL/PostGIS database):

```
python db/import_data.py
python db/qc_geometry.py
python db/fetch_point_covariates.py
python db/rank_sites_priority.py
python db/build_risk_zones.py
python db/run_model_and_store.py "reason for the rerun"
python db/export_geopackage.py
```

POPs (files only, no database needed):

```
python db/fetch_chemicals_tj.py --download-docs
python db/extract_pop_tables.py
python db/build_pop_layer.py
python db/pop_site_covariates.py
python db/pop_screening_model.py          # can be interrupted and resumed
python db/pop_screening_model.py final
python db/pop_child_risk_index.py
python db/pop_data_audit.py
python db/pop_sampling_plan.py
python db/pop_metals_link.py
```

The first step accesses the chemicals.tj portal; the rest work offline.